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A Novel and Optimized Approach for Drilling Operation at Offshore Wellhead Tower Platform Jacket Without Topside Decks to Achieve Significant Advancement in First Oil

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Abstract

WHT consists primarily of two parts, i.e. jacket and topside. When constructing WHT, jacket and topside are installed in a sequential manner. A time gap typically exists between jacket installation and topside installation since it takes longer time to manufacture topside decks than jacket. In order to accelerate the drilling schedule by fully utilizing the time gap, Jacket Drilling, which is drilling with jacket but without topside decks, is sometimes performed between post-jacket installation and pre-topside installation.

In the conventional method, Jacket Drilling operation was divided into two phases. In the first phase, wells were drilled to a certain section through the jacket and then suspended with mud line suspension at the top jacket level until the second phase started. Then, the second phase was carried out after topside deck installation in order to drill the remaining section(s) and complete the wells. scaffolding was erected as a temporary working platform or staircase from the jacket walkway level to the wellhead level, which poses larger risks on the rig crew since it could collapse due to vibration, drop object or clash with other equipment on the rig

On the other hand, in the novel approach that is discussed in this paper, wells are drilled and completed in only one phase by utilizing Drill Deck, which is a permanent platform where rig crew can work around wellhead and BOP in a safer manner without the need to erect extensive scaffolding. After completing the wells by Jacket Drilling, the topside deck is installed on the jacket with Xmas Trees in place.

By adopting the novel method in this paper, it is possible to make significant advancement in the drilling schedule as compared to the conventional method and to provide a safer working environment for the rig crew.

In this paper, key considerations for Jacket Drilling are also elaborated in detail such that the novel method for Jacket Drilling discussed in this paper can be applied to any of the similar WHTs that are newly constructed.

Introduction

It is one of the primary choices to construct WHTs and drill wells by jack-up rig at the area with water depth of approximately less than 100m because such water depth is typically difficult for floater rigs due to smaller operating envelopes from the perspective of allowable rig offset. In the shallower water depth region such as the middle east area, it is quite common to construct offshore WHTs, from which wells are drilled by jack-up rig. In one of the projects in the middle east area, it was planned to construct new WHTs in a brown oil field in order to further develop the field and increase oil production. WHT consists primarily of two parts, i.e. jacket and topside. When constructing WHT, jacket and topside are installed in a sequential manner. In between jacket and topside installation, a time gap (typically 9-12 months) typically exists since it takes longer time to manufacture topside decks than jacket. In order to accelerate the drilling schedule by fully utilizing the time gap, Jacket Drilling, which is drilling with jacket but without topside decks, is sometimes performed between post-jacket installation and pre-topside installation. There are some papers that describe the case study associated with special jack-up rig operation such as subsea Xmas Tree installation by jack-up rig (Abdallah et al. 2023) and jack-up rig installed WHT (Raja et al. 2023). However, there is no paper that elaborates the best approach for planning Jacket Drilling operation although Jacket Drilling itself has been done in many different fields around the world.

Therefore, objectives of this paper are set as follows:

- To explain a novel approach for Jacket Drilling in detail by comparing it with the conventional method
- To elaborate on the technical considerations relevant to the novel Jacket Drilling methodology
- To generalize the key points on Jacket Drilling planning so that the same things discussed in this paper can be applied for any other similar projects around the world.

Comparison between Conventional and Novel Method

In the conventional method (see Figure 1), Jacket Drilling operation was divided into two phases. In the first phase, wells were drilled to a certain section through the jacket and then suspended with mud line suspension at the top jacket level until the second phase started. Then, the second phase was carried out after topside deck installation in order to drill the remaining section(s) and complete the wells. In the second phase, casing strings will be tied back from mud line suspension to wellhead, which will eventually become a weak point from the well integrity perspective.

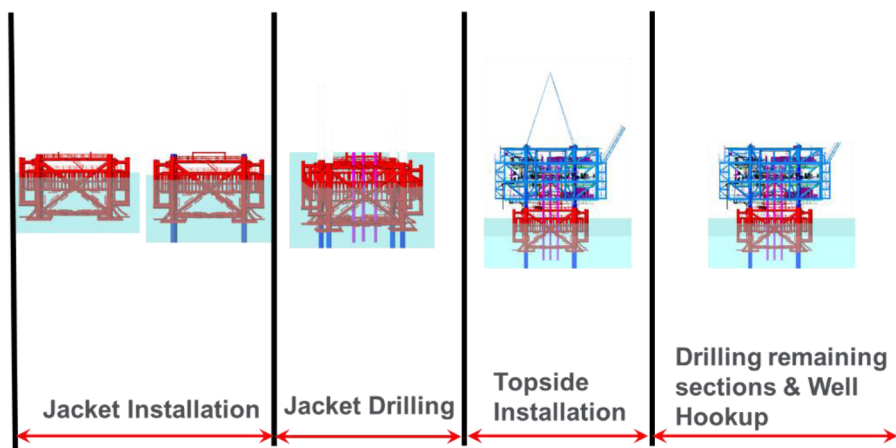


Figure 1—Conventional Method for Jacket Drilling

On the other hand, in the novel approach (see Figure 2), wells are drilled and completed in only one phase by utilizing the specially designed jacket (its design will be detailed in the subsequent section). After completing the wells by Jacket Drilling, the topside deck is installed on the jacket with Xmas Trees in place. As compared to the conventional method, there will be significant advancement in the first oil since wells are drilled and completed in one phase without a second phase, which is also detailed in the subsequent section. Additionally, there is no integrity concern associated with tie-back portion unlike the conventional method.

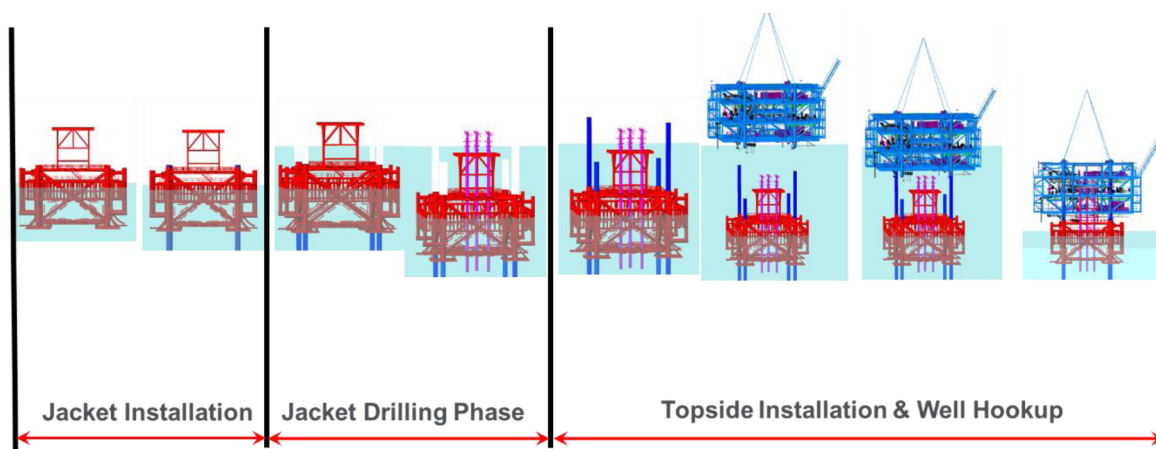


Figure 2—Novel Method for Jacket Drilling

The difference between the two lies mainly in the jacket structure design. Hence, it is vital to work with project team to incorporate the Jacket Drilling concept into the project scope at the early stage of the project (at least from FEED study). Otherwise, it is quite difficult to incorporate the concept after the FEED study.

Details of Jacket Structure

In the conventional way, scaffolding was erected as a temporary working platform or stair case from the jacket walkway level to the wellhead level as shown in Figure 3, which poses larger risks on the rig crew since it could collapse due to vibration, drop object or clash with other equipment on the rig.



Figure 3—Scaffolding erected on jacket

The novel method for Jacket Drilling provides a safer working environment for the rig crew than the conventional method due to Drill Deck (blue portion in Figure 4). Drill Deck is a permanent platform where rig crew can work around wellhead and BOP in a safer manner without the need to erect extensive scaffolding. From the operational perspective, it is pretty similar to the normal operation at WHT other than the fact that working space on Drill Deck is smaller. The dimension of Drill Deck is approximately $8.5\text{m} \times 8.9\text{m}$ as shown in Figure 5. Drill Deck becomes a part of WHT permanent structure after topside deck installation by performing the welding work on the hatched area in Figure 5. Wellhead/Xmas Trees are positioned right above the Drill Deck level. The Drill Deck has a staircase to escape down to the jacket walkway level in an emergency case. This staircase is temporary only and removed before topside decks installation. In addition, the Drill Deck has removable handrails at the topmost deck to avoid man-overboard. The Drill Deck is flush design without anything sticking up above the grating level to remove trip hazards. Each conductor guide on the Drill Deck is equipped with a slot cover to eliminate the possibility of drop object or man-overboard.

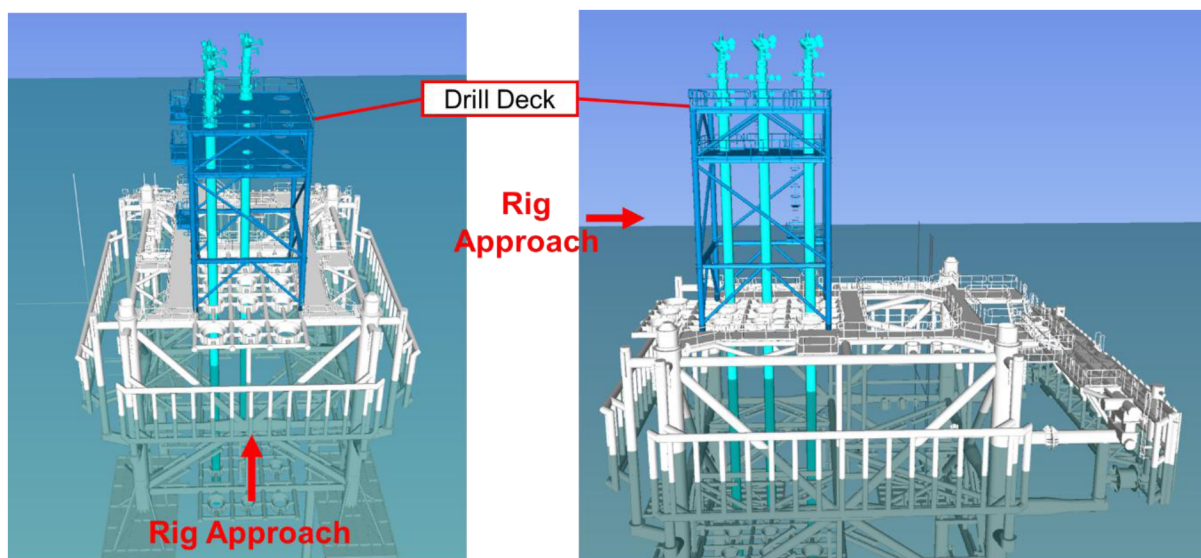


Figure 4—Jacket with Drill Deck (left: looking from Plant South, right: looking from Plant East)

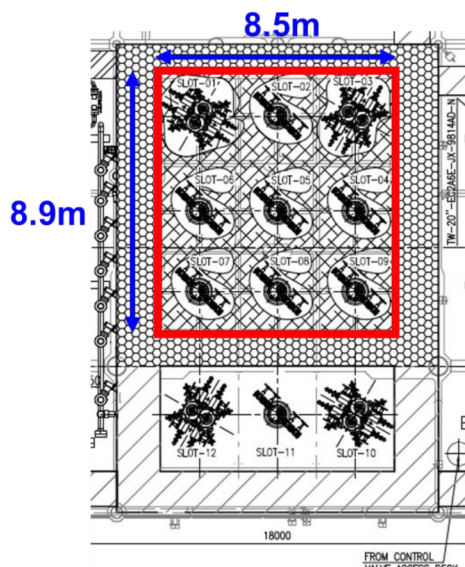


Figure 5—Top view of Drill Deck (marked in red)

Key Considerations for Jacket Drilling

Rig Approach to Jacket

Rig approach is one of the important things to be considered in the planning phase. Accurate rig approach is considerably significant since the rig footprint created on the seabed may remain in place for quite a long years, which may impact the subsequent rig approach. When a jack up rig approaches normal WHT, steel-steel distance (i.e. the distance between rig hull and WHT structure edge) is measured by putting a stick against the rig approach line which is dropped from the structural centerline of the topmost deck before rig approach (although it is sometimes measured by laser). However, such rig approach line cannot be used as a reference in case of Jacket Drilling. If the rig approach line is dropped from the Drill Deck, the distance from the rig to the rig approach line is so large that the distance cannot be measured accurately. Hence, steel-steel distance should be measured against the jacket. In our particular case, it was decided to measure the steel-steel distance against the fender on the rig approach side with clear marking on the center point of the structure as shown in Figure 6. Such marking is also important to align the rig center line and the structure center line.

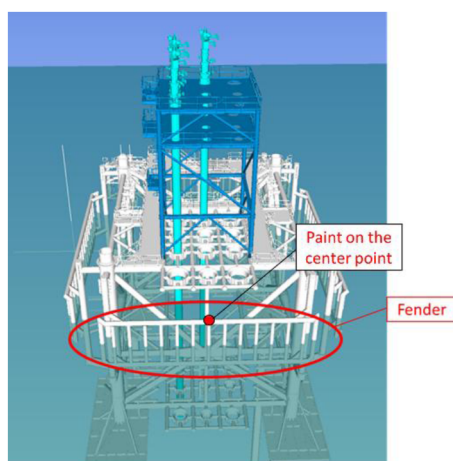


Figure 6—Reference point for measuring steel-steel distance

It is also of great importance to decide the steel-steel distance as if topside decks exist. Namely, steel-steel distance for jacket drilling should be determined such that the rig footprints for Jacket Drilling and drilling after topside installation fall at the same location as each other. Creating multiple different rig footprints on the seabed makes it difficult to perform the subsequent rig approaches. It is best to place spud-cans at the same location every time rig approach is performed. If it is difficult to do so, new spud-can location should be completely away from the existing ones on the seabed. The interaction between a spudcan and a footprint is expected to be minimal when the edge-to-edge distance exceeds one spudcan diameter (SNAME T&R 5-05A, 2008, ISO 19905-1, 2023). Partial overlap between the spudcan and existing rig footprint may cause the leg slide into the existing footprints, which may lead to the rig collapse in the worst case.

Gangway Installation from Rig to WHT, Diverter/BOP Space-out

In our case, there were several different rig designs that were planned to be deployed for Jacket Drilling, i.e. GustoMSC CJ46, KFELS B-Class and KFELS Super B-Class. All of them have different rig designs from each other. In particular, CJ46 does not have a Texas Deck while KFELS B-Class and KFELS Super B-Class have it. In case of CJ46, gangway was planned to be installed from the gangway platform to the Drill Deck (see Figure 7 and Figure 8), which is nothing different from the usual drilling operation at WHT. On the other hand, KFELS B-Class and KFELS Super B-Class do not have such gangway platform, and a

gangway is usually installed from the rig's main deck to WHT. If the gangway is installed from the main deck to the Drill Deck, the inclination of the gangway becomes too steep. Hence, it was decided to install the gangway on the edge of Texas Deck as shown in Figure 9 and Figure 10. In order to fix the gangway on the Texas Deck, some pad eyes were welded as shown in Figure 11. Pad eyes were welded at three different locations so that gangway can be shifted from left/right depending on the actual situation.



Figure 7—Gangway and gangway platform on CJ46

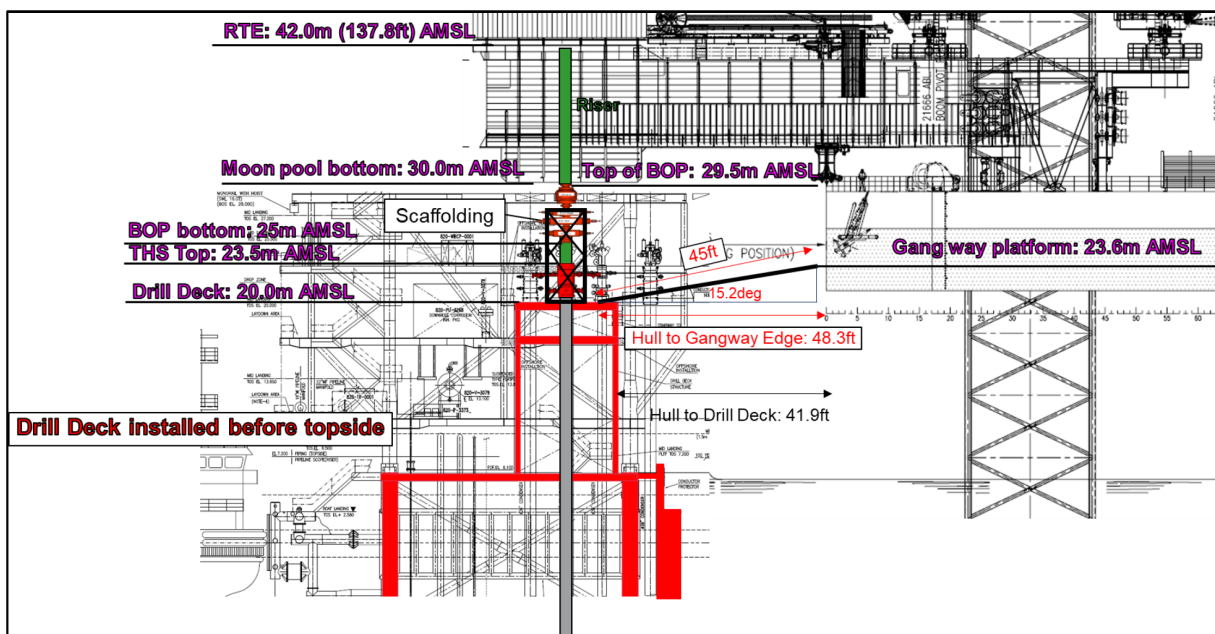


Figure 8—Elevation view of jacket with Drill Deck and rig (CJ46)

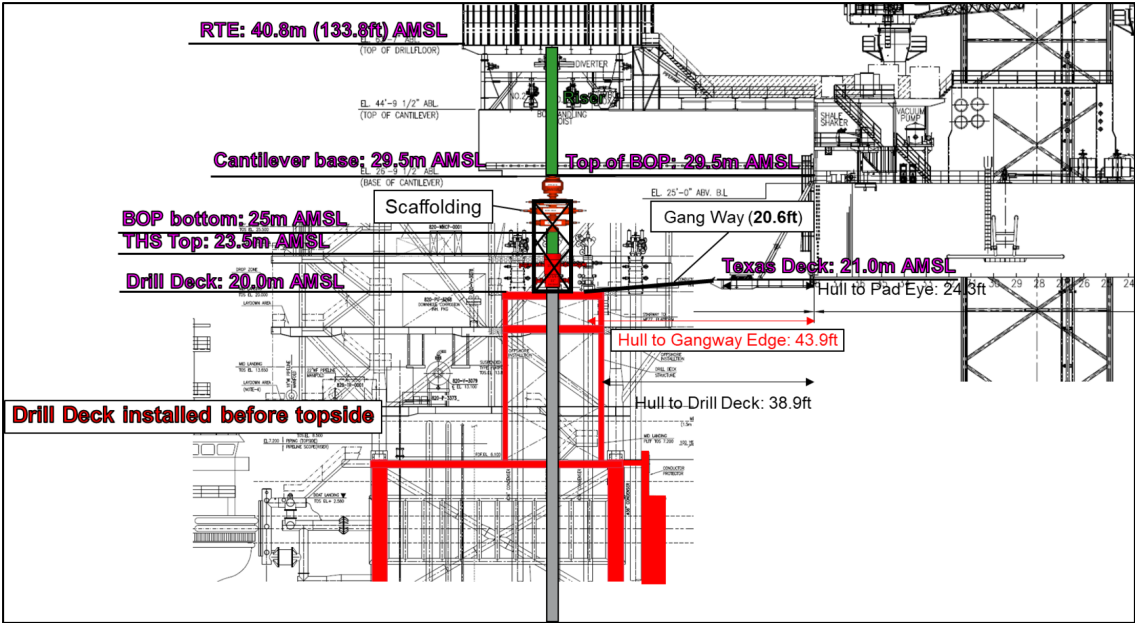


Figure 9—Elevation view of jacket with Drill Deck and rig (KFELS B-Class)

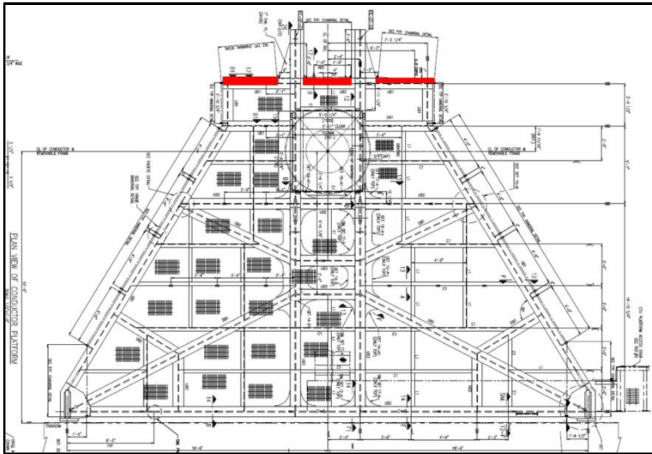


Figure 10—Top view of Texas Deck for KFELS B-Class (gangway to be installed on the red part)



Figure 11—Pad eyes to fix gangway

Diverter/BOP space-out impacts the gangway inclination and its required length. Diverter/BOP space-out was designed not to make the gangway too steep and too long. The length of available riser and bell nipple on the rig must be verified in advance and considered for Diverter/BOP space-out. In case the length of the riser available on the rig is not suitable in terms of Diverter/BOP space-out, fabrication of the new riser and/or bell nipple should be considered.

Above-mentioned case is what was applied to our specific project. But rigs utilized for Jacket Drilling may be different from one project to another. It is imperative to have proper understanding on the rig design details and perform planning sufficiently in advance on how to install gangway from the rig to Drill Deck.

No-go Zone for Topside Installation

It is essential to request an engineering consultant to clearly define the no-go zone for wellhead/Xmas Tree. Figure 12 shows the top view of well bay area, wherein nine (9) slots surrounded by hatched part are the slots available for Jacket Drilling. The hatched part is the no-go zone onto which any of the wellhead/Xmas Tree components should not encroach. This no-go zone is crucial for safe installation of the topside decks. In case any components of wellhead/Xmas Tree intrude into the no-go zone (e.g. wing valve, side outlet valve), it should be temporarily removed before topside installation.

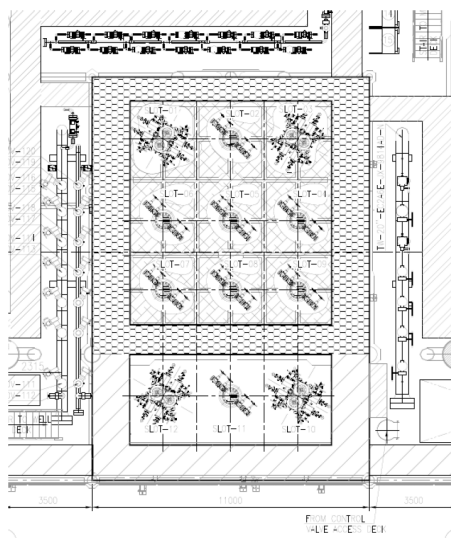


Figure 12—Top view of well bay area

Additional Downhole Barriers

There is a safety measure to avoid the clash between Xmas Tree/Wellhead and topside decks during its installation, e.g. stabbing guide installed on jacket. Nevertheless, it is not possible to completely eliminate the possibility of a clash. In case Xmas Tree/Wellhead is knocked off by the topside decks, the well falls into one-barrier condition, which could escalate to the blowout situation. For an extra precautionary measure, it was planned to have an additional downhole barrier such that two barriers are still maintained even after Xmas Tree/Wellhead is knocked off by topside decks.

Four options were generated as follows:

- Option#1 (Figure 13): Run upper completion with two tubing mounted production packers (barrier disk inside)
- Option#2 (Figure 14): Run intermediate completion and then run upper completion with seal assembly
- Option#3 (Figure 15): Run lower completion with barrier disk and run upper completion

- Option#4 (Figure 16): Run intermediate completion with barrier disk and then run upper completion

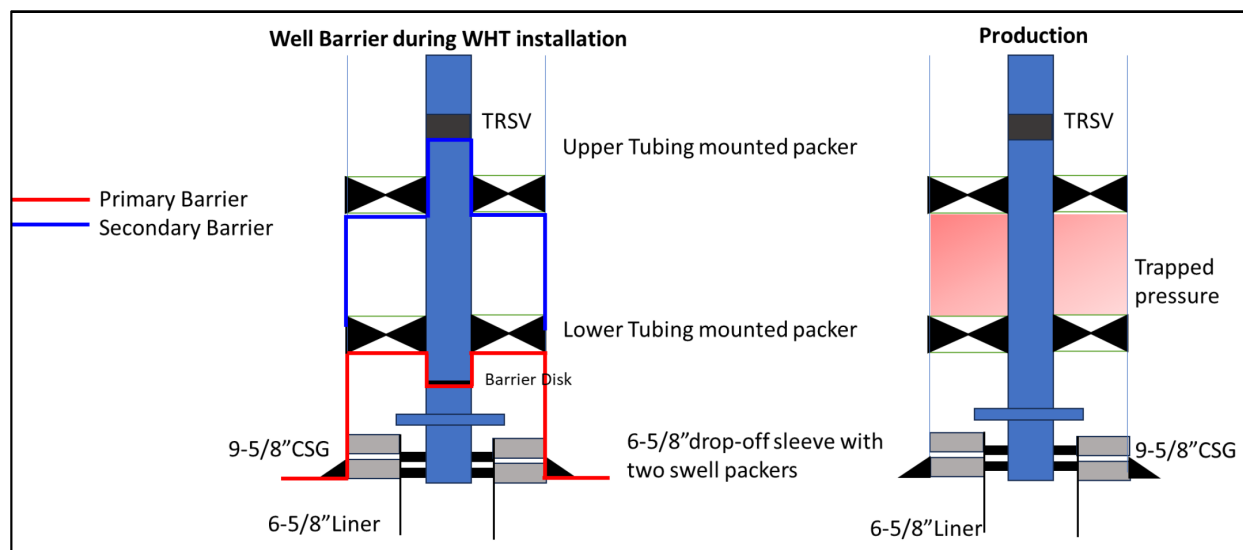


Figure 13—Option#1: Run upper completion with two tubing mounted production packers (barrier disk inside)

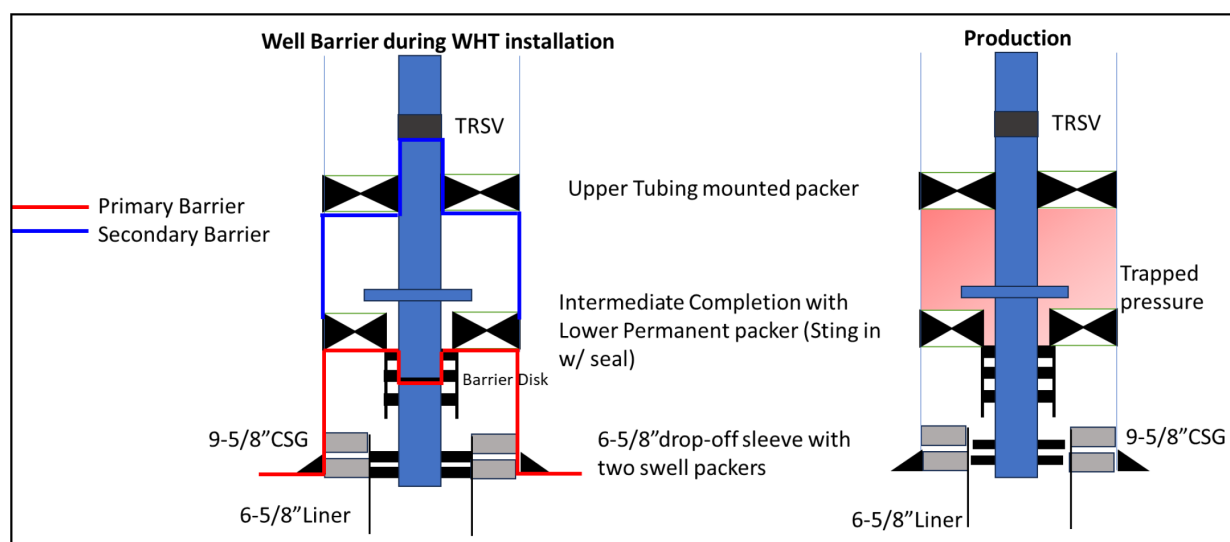


Figure 14—Option#2: Run intermediate completion and then run upper completion with seal assembly

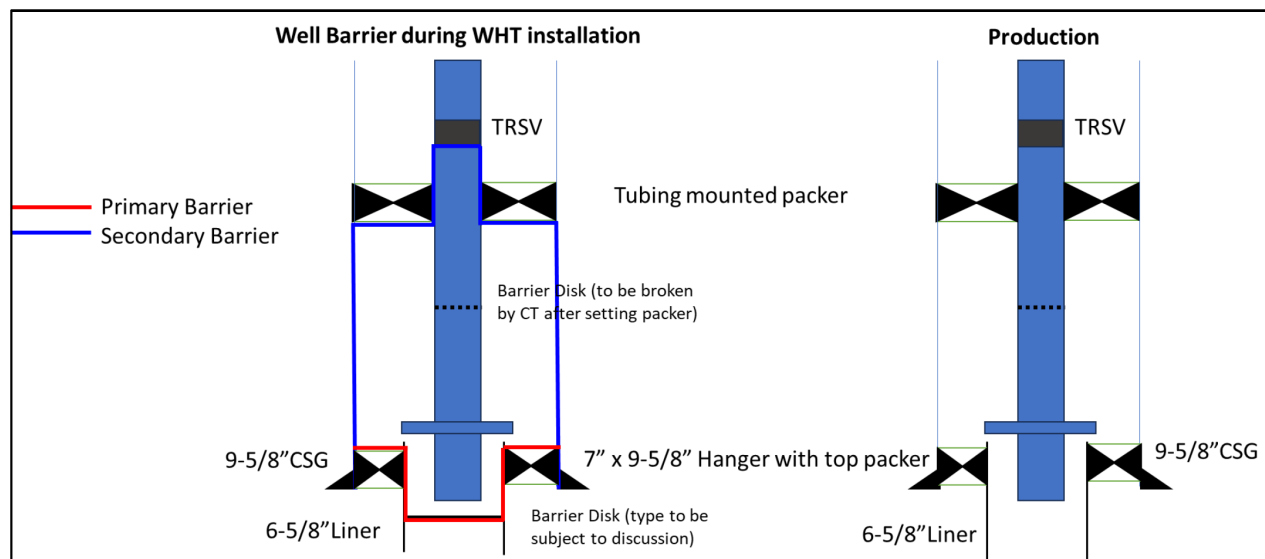


Figure 15—Option#3: Run lower completion with barrier disk and run upper completion

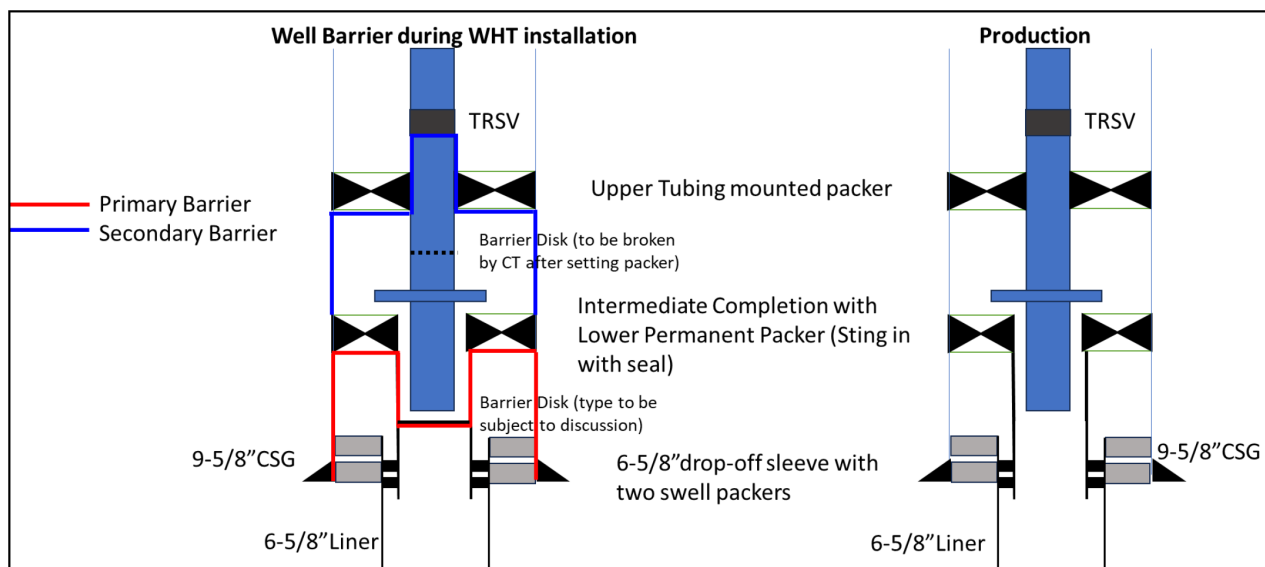


Figure 16—Option#4: Run intermediate completion with barrier disk and then run upper completion

After pros/coms comparison among all the options as shown in Table 1, decision was made to proceed with Option#4. Although it takes additional days, it will be worthwhile implementing double downhole barriers considering the possible consequence, that is, blowout.

Table 1—Pros/Cons Comparison for the populated options

Option#	Pros	Cons
Option#1	- Upper completion can be deployed in one run - Easy to set and recover in the future - Equipment is readily available and part of our standard equipment.	- Requires enough space between production packer to identify the pressure test based on pump volume, however space might be limited due to gas lift mandrel - Unmonitored trap pressure between production packers during production phase (Annulus Fluid Expansion) - Require two different type of packer for setting pressure (Difficult to have sufficient margin for each final setting pressure)
Option#2	Easier to create weak point at seal bore packer to release trapped pressure between packers compared to option#1	- Additional 1.5 days (vs. Option#1) to set intermediate production packer. - Require enough space between production packer to identify the pressure test based on pump volume, however space might be limited due to gas lift - Unmonitored trapped pressure between two production packers during production phase due to annulus fluid expansion - Milling the seal bore packer will generate junk which may damage the lower completion components (ICD, AICD etc.) during workover
Option#3	- No trapped pressure between production packers during production phase. - Easier to pressure-test upper tubing mounted packer	- Additional 2.5 days (vs. Option#1) for coiled tubing operation to break shallower barrier disk on upper completion (pressure-cycle type not to be used since unable to confirm breaking barrier disk by injection). - Difficulty in lower completion installation due to no circulation
Option#4	- No trapped pressure between production packers during production phase. - Easier to pressure-test upper tubing mounted packer	- Additional 4.0 days (vs. Option#1) to set intermediate production packer (1.5days) and coiled tubing operation to break shallower barrier disk on upper completion (2.5days). Pressure-cycle type not to be used since unable to confirm breaking barrier disk by injection. Additional days can be reduced, i.e. 1. Run upper completion w/ set RBP and retrieve it by wireline w/ tractor (may have difficulty in retrieving) or 2. Utilization of wireline tractor + test tool on Nipple - Difficult to confirm upper completion production packer set. - Potential difficulty of breaking only shallower barrier disk

Potential Advancement in First Oil

The biggest advantage of Jacket Drilling is first oil acceleration. Jacket Drilling makes the best use of the time gap between jacket installation and topside installation. Figure 17 shows the comparison of the duration between different scenarios. As compared to "Without Jacket Drilling" and "With Jacket Drilling (Two Phase Drilling)", "With Jacket Drilling (One Phase Drilling)" can make 65 weeks and 35 weeks advancement in first oil schedule respectively (under the assumption of drilling two wells). This will add significant value to the project in terms of economics. In addition to that, there will be notable saving on the rig operation cost because rig operation duration will be remarkably reduced.

However, the important thing to be noted is that the time gap is subject to change driven by jacket and topside fabrication. This means that EPC Contractor's schedule will impact the drilling schedule directly. Therefore, drilling schedule within the time gap should not be too tight and should have some room for delay/acceleration such that EPC Contractor's schedule change can be absorbed.

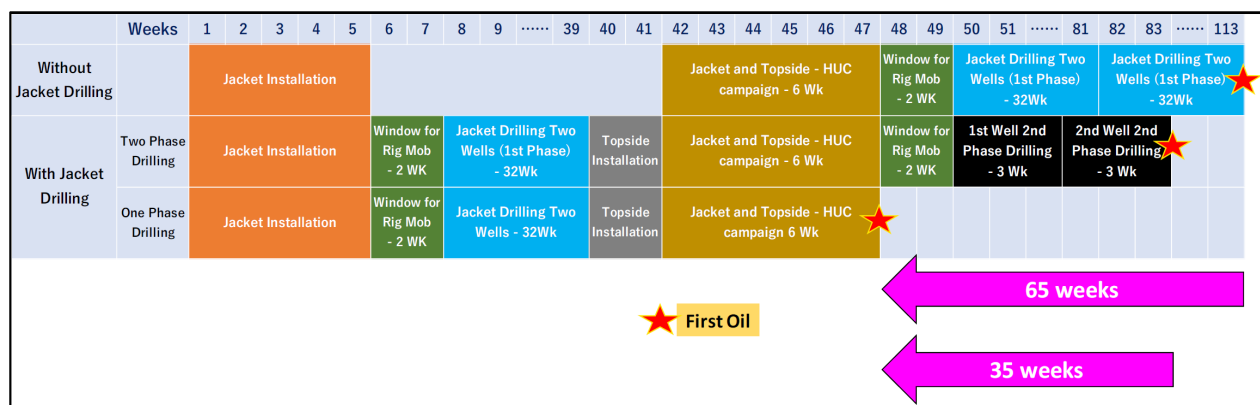


Figure 17—Comparison of the duration between different scenarios

Summary and Conclusion

WHT consists primarily of two parts, i.e. jacket and topside. When constructing WHT, jacket and topside are installed in a sequential manner. A time gap typically exists between jacket installation and topside installation since it takes longer time to manufacture topside decks than jacket. In order to accelerate the drilling schedule by fully utilizing the time gap, Jacket Drilling, which is drilling with jacket but without topside decks, is sometimes performed between post-jacket installation and pre-topside installation. A novel approach for the Jacket Drilling is explained in detail.

In the conventional method, Jacket Drilling operation was divided into two phases. In the first phase, wells were drilled to a certain section through the jacket and then suspended with mud line suspension until the second phase started. Then, the second phase was carried out after topside deck installation in order to drill the remaining section(s) and complete the wells.

On the other hand, in the novel approach, wells are drilled and completed in only one phase by utilizing Drill Deck, which is a permanent platform where rig crew can work around wellhead and BOP in a safer manner without the need to erect extensive scaffolding. After completing the wells by Jacket Drilling, the topside deck is installed on the jacket with Xmas Trees in place.

By adopting the novel method in this paper, it is possible to make significant advancement in the drilling schedule as compared to the conventional method. The novel method for Jacket Drilling discussed in this paper can also be applied to any of the similar WHTs that are newly constructed.

Nomenclature

AICD	Autonomous Inflow Control Device
BOP	Blowout Preventer
EPC	Engineering, Procurement, Construction
FEED	Front End Engineering Design
ICD	Inflow Control Device
RBP	Retrievable Bridge Plug
RTE	Rotary Table Elevation
THS	Tubing Head Spool
TRSV	Tubing Retrievable Safety Valve
WHT	Wellhead Tower

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